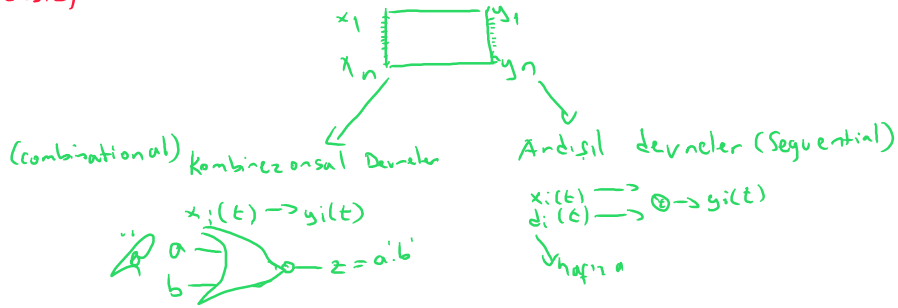


Kombinezonsal Devreler (Statik, Belleksiz)

lojik devreler



A) K.D. Tasarımı

1) Problem Tanıtımı

2) $x_i(t)$, $y_i(t)$ sayıları belirlenir

3) Doğruluk tablosu oluşturulur

4) İndirgeme (cost-effective)

5) Kapı seçimi ve düzenlenmesi

6) Devre lojik şeması çizilir.

2 girişli: 1 çıkışlı devre

girişler eşitse 1 farklıysa 0 verecek



x_1	x_2	z
0	0	1
0	1	0
1	0	0
1	1	1

x_2	x_1	z
0	0	1
0	1	0
1	0	0
1	1	1

$$z = x_1'x_2' + x_1x_2$$



2 bitlik genlik karşılaştırmacı devreyi tasarla



$$A = B \rightarrow z_2 = 1 \quad z_1 = 0 \quad z_0 = 0$$

$$A > B \rightarrow z_1 = 1 \quad z_2 = 0 \quad z_0 = 0$$

$$A < B \rightarrow z_2 = 1 \quad z_1 = 0 \quad z_0 = 0$$

22

a_1	a_0	b_1	b_0	z_2	z_1	z_0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	1	0
0	1	0	1	0	1	0
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	1	0	0
1	0	1	1	1	0	0
1	1	0	0	1	1	0
1	1	0	1	1	1	0
1	1	1	0	1	1	0
1	1	1	1	1	1	0

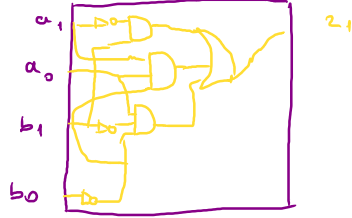
21

a_1	a_0	b_1	b_0	z_2	z_1	z_0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	1	0
0	1	0	1	0	1	0
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	1	0	0
1	0	1	1	1	0	0
1	1	0	0	1	1	0
1	1	0	1	1	1	0
1	1	1	0	1	1	0
1	1	1	1	1	1	0

a_1	a_0	b_1	b_0	z_2	z_1	z_0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	1	0
0	1	0	1	0	1	0
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	1	0	0
1	0	1	1	1	0	0
1	1	0	0	1	1	0
1	1	0	1	1	1	0
1	1	1	0	1	1	0
1	1	1	1	1	1	0

$$Z_3 = \bar{a}_1 b_1 + a_1' a_0' b_0 + a_1 a_0' b_0 + a_1' a_0 b_1 b_0$$

1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0



Kombinasyonel Tümlük Devreler

1) Aritmetik Toplama/Çıkarma devreleri

- yarı toplayıcı
- yarı çıkartıcı
- Tam
- Tam
- Toplama ve çıkarma

2) Seçiciler (Multiplexer) MUX

3) Dağıtıcılar (Demultiplexer) DEMUX

4) Kod Gözütü/Kodlayıcılar (Decoder/Encoder)

5) Aritmetik lojik Birimler (ALU)

6) 7 ?

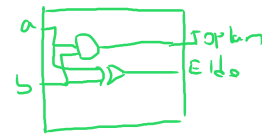
1) Aritmetik topı ve çıkarma devreleri

a) Yarı toplayıcı (half adder)

a	b	Toplam	Elde
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$\rightarrow \text{Elde} = a \cdot b$$

$$\text{Toplam} = a \oplus b$$



b) Tam toplayıcı (Full adder)

Eg	a	b	T	E
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

T için	a	b	c	T	E
0	0	0	1	1	0
0	0	1	1	0	1
0	1	0	1	0	1
0	1	1	0	1	0
1	0	0	0	0	1
1	0	0	1	1	0
1	0	1	0	1	0
1	0	1	1	0	1
1	1	0	0	1	0
1	1	0	1	0	1
1	1	1	0	0	1
1	1	1	1	1	1

$$\rightarrow T = a \oplus b \oplus c$$

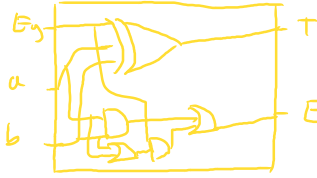
E için

1	0	0	1
1	1	1	1

E için

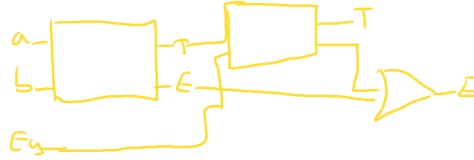
$\backslash a \backslash b$	0	0	1	1
E_3	0	0	1	0
	0	0	1	0
	0	1	1	1

$E = ab + E_3(ab)$



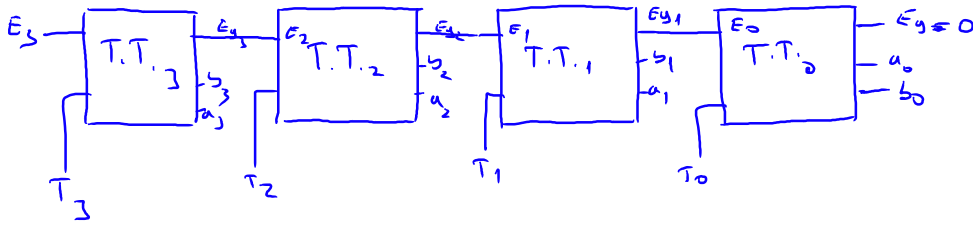
ödev
simüle et

3. Toplayıcı



ödev 2 - Ad Sayı - No. pdf

4 bitlik 2 sayının toplamını bulan paralel toplayıcı devrelerinin tasarımını tam toplayıcılarla yapınız.



c-) Yarı Çıkarsıcı

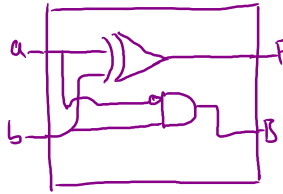
a	b	F_{ek}	Borç
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$F \rightarrow a \oplus b$

$\backslash a \backslash b$	0	1
0	0	1
1	1	0

$B \rightarrow a'b$

$\backslash a \backslash b$	0	1
0	0	0
1	1	0



d.) Tam çıkarsıcı

a	b	Borç	F	B
0	0	0	0	0
0	1	0	1	0
1	0	0	1	0
1	1	1	0	1

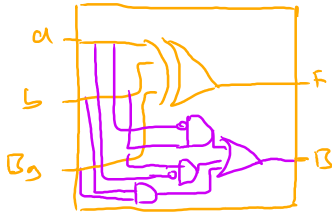
$F = a \oplus b \oplus c$

$\backslash a \backslash b$	0	0	1	1
c	0	1	1	0
	0	1	0	1

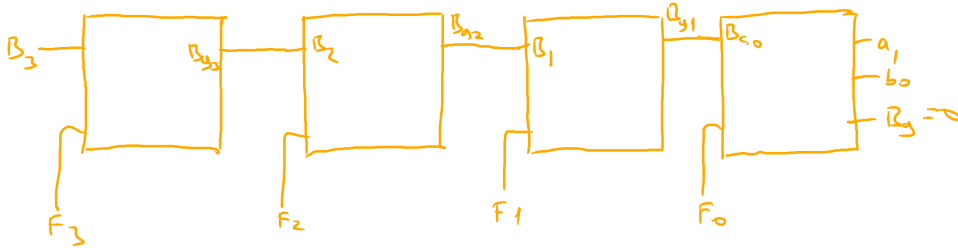
0	0	1	1	1
0	1	0	0	1
0	1	0	0	1
1	0	1	0	0
1	0	1	0	0
1	1	0	1	1

$$\rightarrow a'bB_y + a'bB_y' + a'bB_y + a'bB_y'$$

a b	0	0	1	1
b a	0	0	1	0
	0	0	1	0
	1	1	1	0

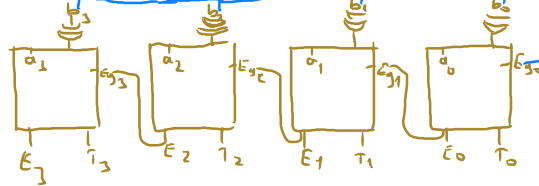
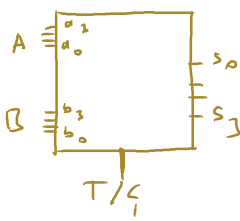
$$\rightarrow a'B_y + a'b + bB_y$$


B₃ dan ödev



2-) Toplama / çıkarma devresi:

4 Bitlik



M=0 → b₀ b₁ b₂ b₃
M=1 → b₀' b₁' b₂' b₃'

2-) Seçiciler (Multiplexer) MUX



S₀ S₁ S₂ S₃
0 0 0 0 → I₀
0 0 0 1 → I₁
1 1 1 1 → I₁₅

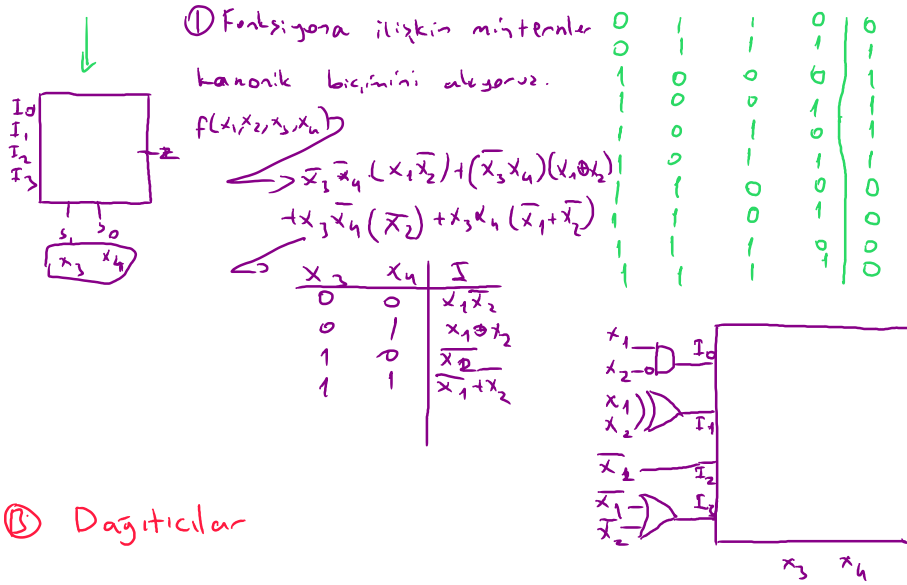
Doğruluk tablosu verilen devreyi x₂ ve x₄ seçme girişleri olacak şekilde 4x1 MUX ile gerçekleştiriniz.

x ₁	x ₂	x ₃	x ₄	f(x ₁ , x ₂ , x ₃ , x ₄)
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0

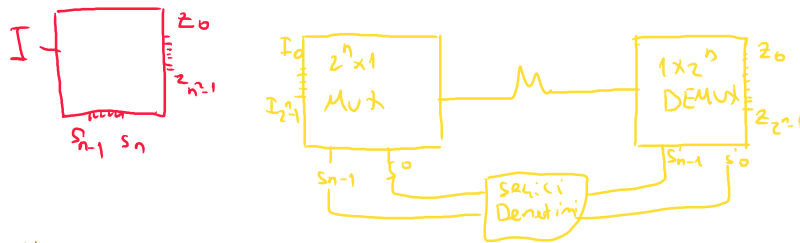
$$f = x_1 \bar{x}_2 + \bar{x}_2 x_3 + \bar{x}_1 x_2 x_4$$

Karnaugh map çizilince

① Fonksiyona ilişkin mintermler



③ Dağıtıcılar



3 bitlik bir ifade 4 kullanıcıya aktarılacaktır.

